

CMPE-172 02

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Term Project M1

Baber Shop and Salon Beauty Appointment Schedule System

I. Problem Scenario

This term project provides an Online Scheduling Appointment system for a Barber and Salon beauty business. The system will allow customers (users) to book appointment for their Hair and beauty service by with the following interactions:

- Browse and select the available time slots
- Select the options of barbers or beauty technicians
- View the appointment details
- Cancel or change the appointment
- Receive notifications about appointments according to date and time

The system will announce to customers if the appointment time/date or barber/technician is not available. It will also prevent double-booking from customers: the system will not allow multiple customers to book the same Barber/Technician with the same date/time appointment. The Barber and Beauty Technician can also interact with the system with these following features to handle the daily business with efficiency:

- Manage their available date and time: add or remove.
- Manage the appointments: view, cancel or reschedule the appointments

The system will send notifications to the customer with the information related to their appointment: Confirmed, Rescheduled, Cancelled.

II. User Roles

- *Customer*: Customer will be able to browse available barber/technician as well as time and date for the appointment, view appointments details and booking history, reschedule or cancel appointments.
- *Barber and Beauty Technician(Service Provider)*: Service Provider will be able to manage their daily or weekly availability, view their appointments details and appointments' history, check appointments as completed or cancelled.
- *Admin*: Admin will be able to manage both Customer and Service Provider accounts, manage the service, prices and overall configuration of the system: delete the account for inactivity, change the account's information as customer or service provider's request...

III. Core Features

- Browse available time slots: Customers will be able to see the available date/time with the filter of Barber/Technician's availability.
- Book appointments: After selecting the time slot, the customer will select the type of service and confirm the appointment. The system will validate the booking appointment by using its booking information.
- Cancel or Reschedule appointments: After confirming the appointment, customers can change the appointment date/time or cancel it completely.

- View appointment history: Both customer and the service provider can view the history of their booking appointments and the upcoming appointments.
- Provider availability management: Service provider (Barber/Technician) can configure their working hours, days of breaks/vacations, add/remove availability.
- Notification: External Mocking Service that sends confirmation about the booking appointment and reminder to the customer via mock email.

IV. Main Entities

Entity	Description	Attributes
user	All user of the system: customer, barber/technician, admin	Username, id, password, email, role, created_at
provider	Barber/Technician data and information	Barber_name, barber_id, tech_name, tech_id, password, avatar, bio, is_active
service	Type of service that the business provides: hair cut, pedicure	Service_name, service_id, description, duration_min, price
availability_slots	Date/time that generate from provider's availability	Slot_id, barber_id, tech_id, slot_date,

		slot_time, start_time, end_time, is_available
appointments	Booked appointment from customer	Appointment_id, customer_id, barber_id, tech_id, slot_id, service_id, status, booked_at, notes

Table 1. The expected main entities of the system

V. System Block Diagram

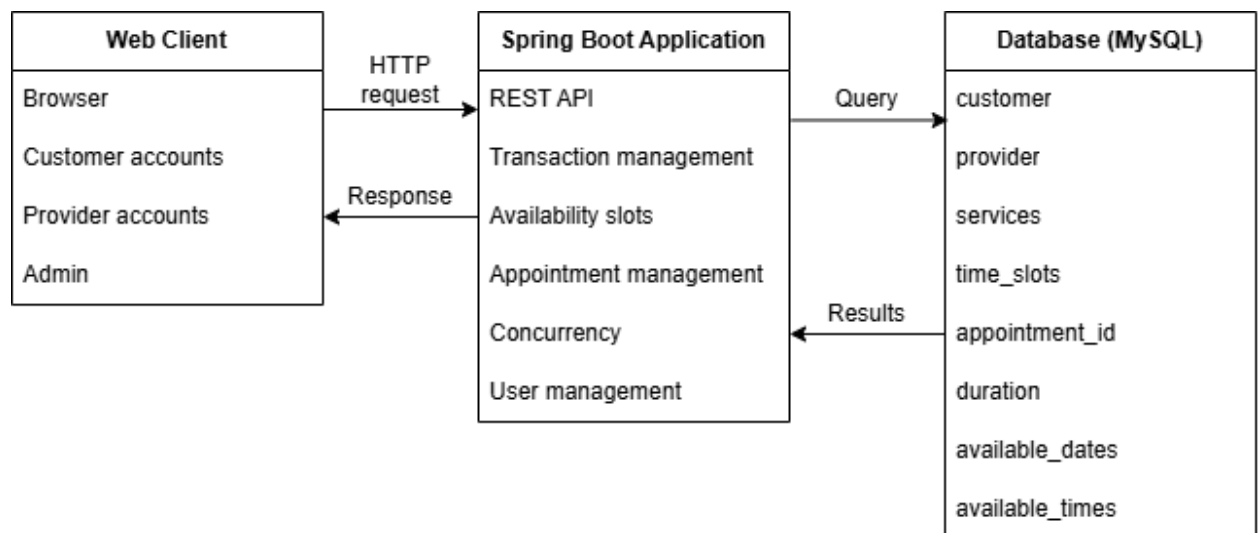


Diagram 1. System Block Diagram

VI. Technology Statement

The project will be implemented using the following technologies:

- Programming Language: Java

- Framework: Spring Boot
- Database: MySQL
- Build tool: Maven
- Architecture:
 - src/main -> controller, model, repository, service